

Suryansh Aryan

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EDUCATION

Kevin T. Crofton Department of Aerospace Engineering, Virginia Tech

Master of Science in Aerospace Engineering, Specialization in Dynamics & Control

Blacksburg, VA

Aug 2023 – Dec 2025

Manipal University

Bachelors of Technology in Aerospace Engineering

Manipal, KN, India

July 2019 – June 2023

EXPERIENCE

Graduate Research Assistant

Space@VT, Virginia Tech

May 2024 – Present

Blacksburg, VA

- Spearheaded NSF-funded project as part of space engineering team, with 2 other teams from NC State University & University of Surrey, driving research solutions in satellite communication resiliency.
- Engineered a real-time mega constellation emulation, establishing 200+ global terrestrial and satellite connections with optimized traffic congestion routing
- Upgraded the satellite node communication resiliency upto 20% through innovative ISL grid scheme and congestion routing, leading to a robust Mininet network for the codebase validation.

UAV Research Intern

Computational Intelligence Lab, Indian Institute of Science

Jan. 2023 – July 2023

Bengaluru, India

- Evaluated 4 key challenges to sustainable precision agriculture via drone technology & constructed a costeffective architecture improving efficiency & affordability for Indian farmers upto \$3000.
- Achieved over \$2,000 cost savings on high-end electronics & reduced maintenance expenses by up to two years.
- Built GNC modules with SPI & CAN-based radars & depth camera drivers for PX4 & K++ flight controllers.
- Improved canopy coverage efficiency by 40%, optimizing variable spraying for crops and trees through ROS/Gazebo simulations.

Guidance-Navigation-Control Intern

General Aeronautics Pvt Ltd

May 2022 – Aug 2022

Bengaluru, India

- Spearheaded the development of scalable SITL & HITL package enhancing trajectory planning for the company's autonomy project, saving \$1000+ on expensive flight controllers & high-resolution sensors.
- Developed 3 innovative algorithms enabling decision-making to detect and navigate obstacles and targets, achieving precision agriculture objectives, saving 3+ months of the control division's efforts.
- Implemented a robust metaheuristic GNC algorithm with APF collision avoidance in SITL, enhancing efficiency by 20%, saving 100+ hours of ground testing.

PROJECTS

Distributed Fault-tolerant Asteroid Inspection | MATLAB, Simulink, Spice

Aug 2024 – Present

- Constructed an SMC controller with stochastic MPC controller to incorporate unmatched uncertainties in asteroid dynamics under frozen conditions.
- Built a swarm reconfiguration strategy that generates optimal satellite-target pairs and compute desired reference trajectories that the adaptive controller takes as an input.

Level-Four Autonomous UAV SITL & HITL Design | Python, Ardupilot, PX4

Aug 2020 – May 2022

- Designed noval Guidance-Control architecture to mitigate realistic unobservable actuator uncertainties existing in low-quality motor and speed controllers.
- Implemented fixed wing/multirotor autonomy by integrating CNN-based Yolov6 object detection and RGBD depth estimation/sensor fusion based localization for Indoor SLAM and outdoor obstacle avoidance applications.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, Julia, Markdown, HTML/CSS

Software Tools: MATLAB/Simulink, Mathematica, Robot Operating System (ROS), Ansys STK, GMAT

Developer Tools: Git, VS Code, Visual Studio, PyCharm, Jupyter notebook, Google colab

Libraries: Spice Toolkit, NumPy, Pandas, Scikit, Matplotlib, SkyField, Tensorflow/Keras, OpenCV

Firmware/Hardware: Ardupilot, PX4, Betaflight, Navio2, Raspberry Pi 4, Jetson Nano